$$\cos\alpha' = \frac{\sqrt{1-V^2/c^2}\cos\alpha}{1-V\sin\alpha/c} \approx \cos\alpha\left(1+V\sin\alpha/c\right)$$
$$\sin\alpha' = \frac{\sin\alpha - V/c}{1-V\sin\alpha/c} \approx \sin\alpha - V\cos\alpha^2/c\,. \quad (38)$$

$$\sin\alpha' = n\sin\beta', \quad (39)$$

$$v_{ph} = \frac{c}{n}. \quad (40)$$

(36) (37)

$$v_m \sin\beta = \frac{v_{ph}\sin\beta' + V}{1+\dfrac{v_{ph}V\sin\beta'}{c^2}} \approx v_{ph}\sin\beta' + V\,, \quad (41)$$

$$v_m \cos\beta = \frac{\sqrt{1-V^2/c^2}}{1+\dfrac{v_{ph}V\sin\beta'}{c^2}} v_{ph}\cos\beta' \approx v_{ph}\cos\beta'\,. \quad (42)$$

(38)-(42)

$$\sin\beta \approx \frac{1}{n}\sin\alpha - \frac{n^2+\cos 2\alpha}{n}\frac{V}{c}, \quad (43)$$

$$A_1 = \frac{1}{n}\sin\alpha, \quad (44)$$

$$B_1 = -\frac{n^2+\cos 2\alpha}{n}. \quad (45)$$

3.3 **[0,4점]** (38)-(42)

$$v_m \approx \frac{c}{n} + V\left(1-\frac{1}{n^2}\right)\sin\beta\,. \quad (46)$$

$$A_2 = \frac{c}{n}, \quad (47)$$

$$B_2 = \left(1-\frac{1}{n^2}\right)\sin\beta\,. \quad (48)$$

3.4 **[0,9점]** , β
(48) $\pi/2$,

$$v_+ = \frac{c}{n} + V\left(1-\frac{1}{n^2}\right), \quad (49)$$

$$v_- = \frac{c}{n} - V\left(1-\frac{1}{n^2}\right). \quad (50)$$

$2L$,
Δt

$$\Delta t = \frac{2L}{v_-} - \frac{2L}{v_+} \approx \frac{4Lv\left(n^2-1\right)}{c^2}, \quad (51)$$